ĐORĐE MARKOVIĆ

Full Stack & AI Engineer

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Full-stack engineer focused on AI products, building Node.js backends since 2022, from team internships to products I now run in production. I work across the stack: API and data design, React frontends, LLM integration backed by evals, payments, testing and deployment. I also maintain Vir, an open-source knowledge tool for coding agents, and use AI coding tools inside a reviewed, test-first workflow.

TECHNICAL SKILLS

AI engineering: Anthropic, OpenAI and Gemini APIs (tool use, structured outputs, streaming), Whisper, vision models, multi-vendor routing with quality gates and failover, LLM evals (LLM-as-judge with calibration, bootstrap and Wilson confidence intervals, lm-eval), MCP servers, prompt-injection hardening, cost tracking; AI coding tools (Claude Code: skills, hooks, sub-agents)

Frontend: React, TypeScript, Next.js, Vite, MUI, Tailwind, Zustand, React Hook Form, PWAs (IndexedDB, web push, offline-first), React Native (Expo)

Backend & data: Node.js, Express, Python, MongoDB (Mongoose, transactions, GridFS), PostgreSQL/Supabase (RLS, SQL functions, pg_cron, Edge Functions), SQLite, Zod, yup, SSE streaming, JWT refresh rotation and revocation, OAuth, Swagger/OpenAPI, SendGrid, cron jobs

Payments: OTP Bank card gateway (HMAC, signed XML API, settlement, reconciliation), PayPal, Paddle

Testing & delivery: Vitest, Playwright (incl. live smoke tests against production), Cypress, node:test, Jest, pytest, GitHub Actions (tag-triggered releases with provenance), Vercel, Render, Cloudflare, Heroku, npm publishing

EXPERIENCE

Full Stack Engineer · GrowthQ Lab DOO (co-founded software studio) Aug 2025 - Present

- Build and run SaaS products end to end, owning architecture, frontend, backend, AI integrations, billing, automated testing and production delivery on Vercel and Render.
- Work with AI coding tools inside a reviewed workflow: shared project conventions, tests, code review and an automated pre-release audit before every deploy.
- Hardened AI features for production: per-user rate limiting that fails closed, per-document LLM cost tracking with fallback-rate alerts, shared Zod contracts between client and API, token revocation and OAuth state nonce.
- Built an offline-first asset-inventory PWA for a government client: QR scanning (native BarcodeDetector with zxing fallback), append-only scan log with pure reconciliation, automatic pre-change backups in a separate IndexedDB with validated restore; ran a security review and fixed 26 of its 27 findings; 175 tests.

Freelance Full Stack Developer · Remote, alongside GrowthQ Lab Nov 2025 - Apr 2026

- Built two client web apps in a two-person team, working across frontend and backend and through client feedback rounds.
- Sports-court booking PWA (Next.js, TypeScript): Google and Facebook OAuth login, installable PWA with web push notifications, and court location and reservation flows.
- Multilingual website for a cultural organization: React/Vite frontend with language switching, Java backend on MongoDB, and event, workshop and gallery pages.

Data Engineering Trainee · Functor (remote) Jul 2025 - Sep 2025

- Remote training program in data engineering: building data pipelines with Docker, Apache Spark, Kafka and Airflow.

Freelance Web Developer · Gracanica Innovation Centre May 2024 - Aug 2024

- Designed the initial look of the organization's website and built it in React: course, event, project, team and partner pages, plus a newsletter signup; handed it over to the team, who later rebuilt it in Webflow for easier editing.

Internship Trainee · Reputeo Feb 2024 - Mar 2024

- Built web crawlers and data extraction workflows with Python and Scrapy.

Backend Developer · ReservEat Mar 2023 - Jun 2023

- Built the authentication API for a booking platform (Node.js, Express, TypeScript, MongoDB): registration with duplicate-account checks, JWT login, email-confirmation reminders and password reset, with SendGrid email templates and yup validation.
- Built the booker, profile and request APIs: profile onboarding, booker updates, and booking requests listed with their offer counts, with responses shaped so credentials never leave the server.
- Extended the MongoDB data models for users and bookers with Mongoose schemas and matching yup validation; documented every endpoint in Swagger/OpenAPI.
- Worked with the frontend developer on the React client: responsive navigation and the password-reset screen, branding and typography updates, and Cypress end-to-end tests for the main user flows.
- Set up and ran the API on Heroku; worked in a Scrum team with ticketed tasks, every change shipped through a reviewed pull request.

Back End Developer Intern · 10minutendeutsch Jul 2022 - Jan 2023

- Built Node.js, Express and MongoDB REST APIs for a German-learning platform: password reset with token links, profile updates with yup validation, and user search with filters, status and pagination.
- Built the backend for the German level tests (answer keys per test) and extended the user model with basic and advanced account tiers.
- Built transactional email with SendGrid dynamic templates in the user's language, used across sign-up, password reset, the waiting list and the contact form; waiting-list signup supports subscription types.

- Designed the conversation-match data model and wrote a weekly cron job that pairs learners by level and interests into week-long conversation matches.
- Wrote Cypress tests, tested APIs in Postman and ran the backend on Heroku; every PR was reviewed by senior developers.

SELECTED PROJECTS

Pripremi.rs · AI SaaS for Serbian teachers 2026 - Present

React, Vite, MUI · Express, TypeScript, MongoDB · OpenAI, Gemini · Vercel, Render · pripremi.rs

- Built and run solo a lesson-plan generator for Serbian teachers that turns a short brief into a Word-ready preparation in about 2 minutes; live in production with organic users, grown without paid marketing.
- Multi-vendor LLM orchestrator (OpenAI GPT + Gemini) with a runtime quality gate, quality fallback, vendor failover, per-document cost tracking, a rolling fallback-rate alert and a post-deploy LLM canary.
- In-house eval harness over the production pipeline: OpenAI, Gemini, DeepSeek and Claude arms, token economics per variant, LLM-as-judge scoring calibrated with planted defects, and Wilson confidence intervals; production models were switched on cost per valid document, and a re-test against the newest Claude models kept the cheaper model in production.
- "Moj obrazac": teachers upload their own lesson-plan template and an LLM extracts its structure, with the extraction model chosen by an eval on public lesson plans.
- OTP Bank card payments: HMAC-SHA256 MAC pinned to spec vectors, signed redirect and XML API with fail-unknown handling, a payment ledger with exactly-once settlement, and a 15-minute reconciliation job.
- Security pass: access-token revocation, hashed refresh tokens, OAuth state nonce, CSRF header, trusted client IP behind Cloudflare, atomic trial admission and a quota limiter that fails closed.
- 2,700+ hand-written tests across node:test, Playwright and Vitest, plus a live smoke test of every main flow against production.

Vir · Open-source knowledge layer for coding agents 2026 - Present

TypeScript, Node.js, SQLite, MCP · virwiki.dev · npm: @djolex999/vir-cli · MIT

- CLI that distills Claude Code and Codex sessions (plus articles and PDFs) into a searchable knowledge base, served back to agents through an MCP server and managed CLAUDE.md/AGENTS.md blocks. Listed on awesome-claude-code.
- Two-model pipeline (Haiku classifies, Sonnet distills) behind free heuristic filters, so paid calls are gated by design; agent-CLI calls run sandboxed (no tools, strict MCP config) because transcripts are untrusted input.
- *vir connect*: clusters lessons that recur across 3+ sessions (union-find), requires verbatim quotes from cited notes, and proposes agent rules for human approval.
- Retrieval eval harness with paired-bootstrap confidence intervals; its results switched off a shipped reranking default that hurt search quality. 1,400+ tests, 60+ npm releases, ~24k lines of production TypeScript, 20 GitHub stars.
- Companion Obsidian plugin, listed in the Obsidian community directory with 180+ downloads: review queue in the sidebar, contract tests that pin the CLI's JSON output, tag-triggered releases with build provenance.

GrowthQ · AI brand content engine 2025 - Present

React, Vite · Express, TypeScript, MongoDB · pnpm monorepo · OpenAI, Kie.ai, Cloudinary · growthq.rs

- SaaS that turns brand profiles into branded social content, with shared Zod API contracts between client and server and 550+ tests.
- AI onboarding: brand inference from images (OpenAI vision), Whisper voice input and a free AI-generated logo; Kie.ai image generation on an atomic credit system with refunds on every failure path.
- PayPal capture inside a MongoDB transaction that only settles pending payments; SSRF-safe image references. Investor and market feedback drove a pivot to a US-market version (USD plans, post scheduling, web push, AI video reels with Remotion and ElevenLabs), in development.

serbeval · Public Serbian LLM benchmark 2026

Python, lm-evaluation-harness, pytest · github.com/djolex999/serbeval · Hugging Face dataset

- Built a public benchmark of how well LLMs write Serbian: 351-item official set (grammar, verb forms, ijekavian, orthography), answers checked against Pravopis 2010, 30 models evaluated through Anthropic, OpenRouter and Kie adapters; published on Hugging Face, with an upstream PR opened to EleutherAI lm-evaluation-harness.
- When top models saturated v1 at about 98%, shipped v2 after public feedback from Aleksa Gordić: 76 verified whole-passage correction tasks run on 19 models, raising frontier error rates 5 to 10 times with a stable ranking (Spearman 0.996). Also caught an answer-position bias that let a constant answer score 80%; 490+ pytest tests.

Other projects

- **sterija** (live): bilingual Astro site on Cloudflare Workers with "Maru", a Claude-powered chat assistant with worst-case cost reservation, a per-IP daily budget, rate limiting and an eval suite; 250+ tests.
- **angler** (experimental): photo-culling tool I am building for The Next Angle, a photo and video studio I invest in. A Python CLI scores shots on faces, sharpness, exposure and bursts and writes Lightroom ratings via XMP; validated on a 108-photo real shoot, 450+ tests, CI on Windows and macOS.

EDUCATION

BSc Computer Science · Faculty of Natural Sciences, North Mitrovica · 2019-2026 October 2026

Thesis: “User Experience 3.0: Applying Artificial Intelligence on Web Platforms”.

ADDITIONAL

Awards: Launcher.rs winner (2025); InnoFuture Hackathon, 1st place (2024), co-creator of FarmLink, a digital marketplace for local producers; Arduino Makers League, 2nd place (2024).

Events: pitched GrowthQ to investors at Web Summit, Lisbon (2025) and GITEX Europe, Berlin (2026).

Mentorship: mentored by Vukota Pekovic (Cisco Silicon One); Mentor the Young mentee. **Languages:** Serbian, English (professional).